

Speech in NLP: Speech-to-Text (STT / ASR)

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1. Introduction: ASR goals, history, and the arc from classical to deep learning
2. Classical ASR: noisy channel model, WER, and HMM–GMM architecture
3. Acoustic features and training: MFCCs, mel filterbanks, GMMs, and Viterbi alignment
4. Hybrid HMM–DNN ASR: neural acoustic modeling and recurrent hybrids
5. End-to-end ASR: Connectionist Temporal Classification (CTC) and RNN-Transducer (RNN-T)
6. Self-supervised learning: Wav2Vec 2.0
7. Modern ASR: Conformer, SpecAugment, and Whisper
8. Streaming ASR: causal and chunked encoders, RNN-T, and endpointing
9. Deployment, fairness, benchmarks, and references

- ▶ **Broad goal.** Given audio, produce exact transcript of words from speaker(s).
- ▶ **Verbatim (Exact) Transcription:** Capturing every “um,” word fragment, repetition, and disfluency to preserve the true speech signal.
- ▶ **Clean (Readable) Transcripts:** Text that omits or normalizes minor speech errors and filler words, yielding more fluent, reader-friendly output.
- ▶ **Real-Time, Low-Latency Operation:** For interactive settings (voice assistants, live captions), ASR must process audio as it arrives. Keep end-to-end latency under 200 ms for natural dialogue flow.
- ▶ **On-Device and Edge Constraints:** Need to run ASR on devices with no cloud support. Compute and memory may be limited.

Switchboard corpus conversation example

► In 2010:

- ASR, TTS, dialog all used specialized, hard-to-build modeling approaches
- Industry application of SLU systems limited. ASR “didn’t quite work well enough”

► Today:

- ASR, TTS, dialog all use deep learning approaches. Less specialized and better performance
- Spoken language systems are everywhere!
- New tools enable building full systems

Automaton:

- ▶ Markov 1911
- ▶ Turing 1936
- ▶ McCulloch-Pitts neuron (1943)
 - mar.bsee.swin.edu.au/cht/net204/lecture10/ann/node1.html
 - www.epfl.ch/mtap/tutorial/english/mcpitts.html
- ▶ Shannon (1948) link between automata and Markov models

Human speech processing

- ▶ Fletcher at Bell Labs (1920's)

Probabilistic/Information-theoretic models

- ▶ Shannon (1948)



Radio Rex toy and box (c. 1920s)

1920's Radio Rex

- ▶ Celluloid dog with iron base held within house by electromagnet against force of spring
- ▶ Current to magnet flowed through bridge which was sensitive to energy at 500 Hz
- ▶ 500 Hz energy caused bridge to vibrate, interrupting current, making dog spring forward
- ▶ The sound “e” (ARPAbet [eh]) in Rex has 500 Hz component

- ▶ **1952: Bell Labs single-speaker digit recognizer**
 - Measured energy from two bands (formants)
 - Built with analog electrical components
 - 2% error rate for single speaker, isolated digits
- ▶ **1958:** Dudley built classifier that used continuous spectrum rather than just formants
- ▶ **1959:** Denes ASR combining grammar and acoustic probability

► Hidden Markov Model 1972

- Independent application of Baker (CMU) and Jelinek/Bahl/Mercer lab (IBM) following work of Baum and colleagues at IDA

► ARPA project 1971–1976

- 5-year speech understanding project: 1000 word vocab, continuous speech, multi-speaker
- SDC, CMU, BBN
- Only 1 CMU system achieved goal

► 1980's +

- Annual ARPA “Bakeoffs”
- Large corpus collection:
 - TIMIT
 - Resource Management
 - Wall Street Journal

Hub5'00 Evaluation (Switchboard / CallHome)

(Possibly trained on more data than SWB, but test set = full Hub5)

WER (SWB)	WER (CH)	Paper	Published
4.9%	9.5%	An investigation of phone-based subword units for end-to-end speech recognition	April 2020
5.0%	9.1%	The CAPIO 2017 Conversational Speech Recognition System	December 2017
5.1%	9.9%	Language Modeling with Highway LSTM	September 2017
5.1%		The Microsoft 2017 Conversational Speech Recognition System	August 2017

WSJ

(Possibly trained on more data than WSJ.)

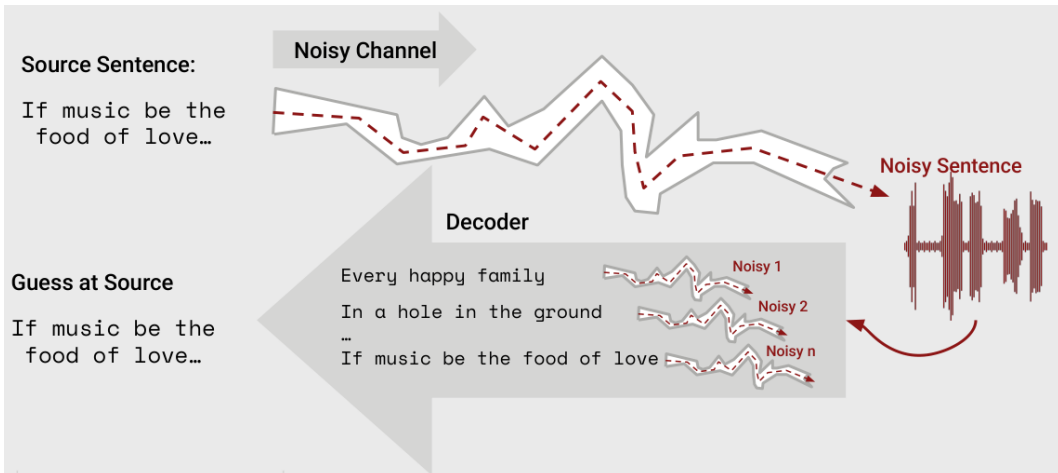
WER eval'92	WER eval'93	Paper	Published
5.03%	8.08%	Humans Deep Speech 2: End-to-End Speech Recognition in English and Mandarin	December 2015
2.9%		End-to-end Speech Recognition Using Lattice-Free MMI	September 2018
3.10%		Deep Speech 2: End-to-End Speech Recognition in English and Mandarin	December 2015

LibriSpeech

(Possibly trained on more data than LibriSpeech.)

WER test-clean	WER test-other	Paper	Published
5.83%	12.69%	Humans Deep Speech 2: End-to-End Speech Recognition in English and Mandarin	December 2015
1.9%	3.9%	Conformer: Convolution-augmented Transformer for Speech Recognition	May 2020
1.9%	4.1%	ContextNet: Improving Convolutional Neural Networks for Automatic Speech Recognition with Global Context	May 2020

- ▶ Search through space of all possible sentences.
- ▶ Pick the one that is most probable given the waveform.



- ▶ What is the most likely sentence out of all sentences in the language L that generated some given **acoustic input** O ?
- ▶ Treat acoustic input O as sequence of individual observations:

$$O = o_1, o_2, \dots, o_T$$

- ▶ Define a sentence as a sequence of words:

$$W = w_1, w_2, \dots, w_n$$

- **Probabilistic implication:** Pick the highest probability word sequence W :

$$\hat{W} = \arg \max_{W \in L} P(W | O)$$

- We can use Bayes' rule to rewrite this:

$$\hat{W} = \arg \max_{W \in L} \frac{P(O | W)P(W)}{P(O)}$$

- Since denominator is the same for each candidate sentence W , we can ignore it for the argmax:

$$\hat{W} = \arg \max_{W \in L} P(O | W)P(W)$$

Likelihood



Prior



$$\hat{W} = \operatorname{argmax}_{W \in L} P(O|W)P(W)$$

$$\text{Word Error Rate} = 100 \times \frac{\text{Insertions} + \text{Substitutions} + \text{Deletions}}{\text{Total Words in Correct Transcript}}$$

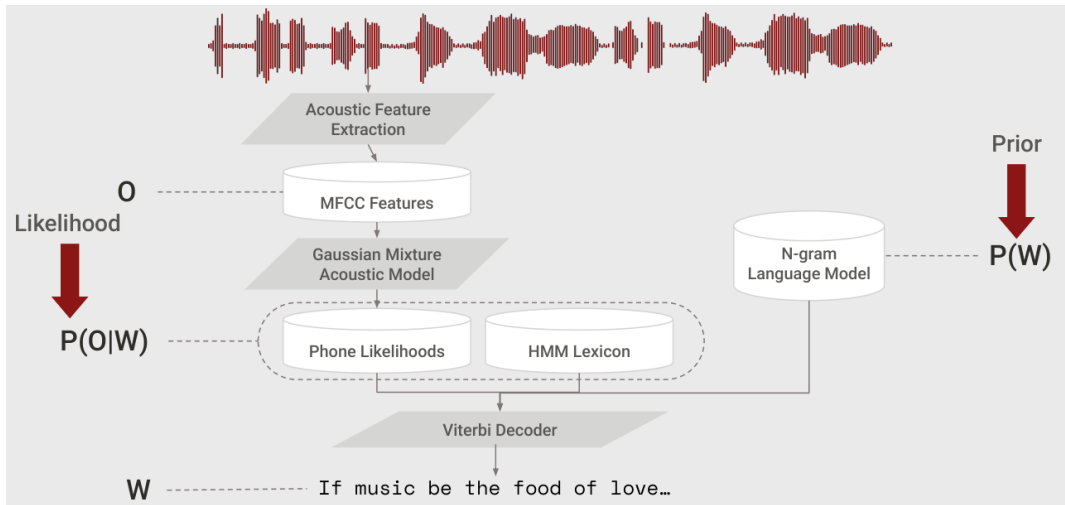
Can be >100%.

Doesn't distinguish between function words (of, they, he, she) and more important content words

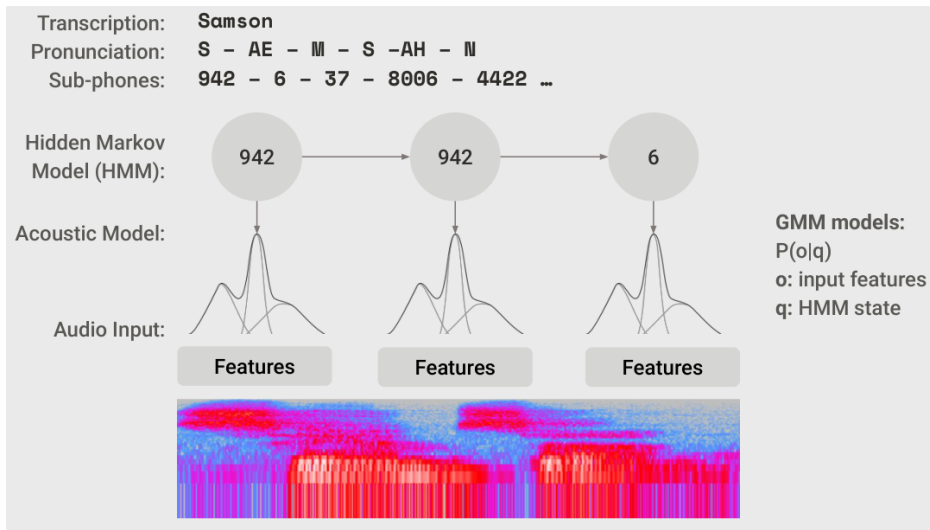
Compute best alignment of reference and hypothesis to count errors:

REF:	i	***	**	UM	the	PHONE	IS		i	LEFT	THE	portable	****	PHONE	UPSTAIRS	last	night	
HYP:	i	GOT	IT	TO	the	*****	FULLEST	i	LOVE	TO		portable	FORM	OF		STORES	last	night
Eval:	I	I	S		D	S		S	S				I	S		S		

Speech Recognition Architecture



Classical ASR:HMM-GMM system



► **Markov assumption:**

$$P(q_t \mid q_1, \dots, q_{t-1}) = P(q_t \mid q_{t-1})$$

► **Output-independence assumption:**

$$P(o_t \mid O_1^{t-1}, q_1^t) = P(o_t \mid q_t)$$

$Q = q_1, \dots, q_T$ a set of N states
 $A = a_{ij}$: **transition probability matrix** A ,
each a_{ij} representing the
probability of moving from state i
to state j . $\sum_j a_{ij} = 1 \ \forall i$
 $O = o_1, \dots, o_T$ a sequence of T observations, each
one drawn from a vocabulary
 $V = v_1, \dots, v_W$
 $B = b_j(o_t)$ a sequence of **observation
likelihoods**, also called *emission
probabilities*, each expressing the
probability of an observation o_t
being generated from a state j
 q_0, q_F a special **start state** and (**final**)
end state that are not associated
with observations, together with
transition probabilities $a_{q_0 i}$ out of
the start state and $a_{j q_F}$ into the
end state

► Feature Extraction:

- 39 MFCC features

► Acoustic Model:

- Gaussian mixture models (GMM) to approximate $p(o|q)$

► Lexicon/Pronunciation Model:

- HMM: which phones can follow each other. Sequence constraints from phonetic lexicon

► Language Model:

- N-grams for computing $p(w_i|w_{i-1})$

► Decoder:

- Viterbi algorithm: dynamic programming for combining all pieces to estimate a word sequence from audio

Source: Stanford CS224S, Lecture 9 (2025 Spring), slide 15 (adapted).

You are a climatologist in the year 2799 studying global warming trends

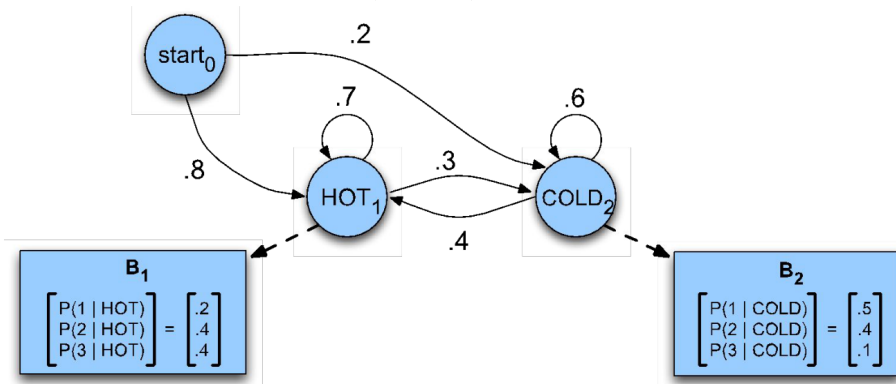
- ▶ You can't find any records of the weather in Baltimore, MD for summer of 2008, but you find Jason Eisner's diary! Which lists how many ice-creams Jason ate every date that summer

Our job: Infer the sequence of daily temperatures (HOT vs COLD)

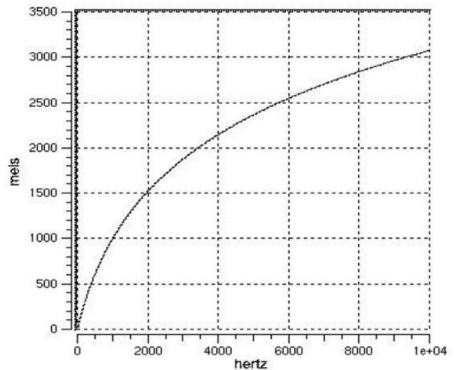
- We do not observe temperature. We observe number of ice creams eaten
- There is a higher probability of eating more ice cream on hot vs cold days
- There is some relationship of next day temperature given current day temperature

Simple HMM: Estimating climate from ice cream observations

Estimate the unobserved state daily_temp (HOT, COLD) based on observations of ice creams consumed

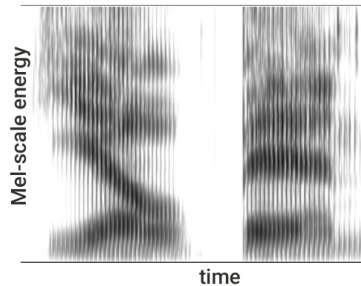
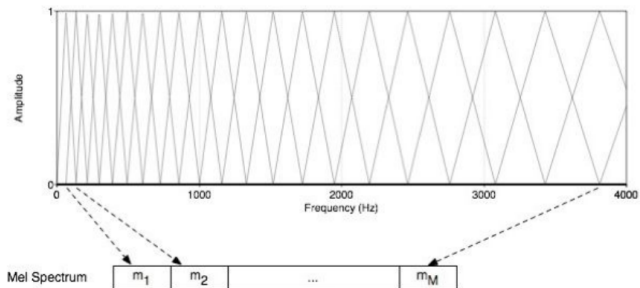


- ▶ Human hearing is not equally sensitive to all frequency bands
- ▶ Less sensitive at higher frequencies, >1000 Hz
- ▶ I.e. human perception of frequency is non-linear

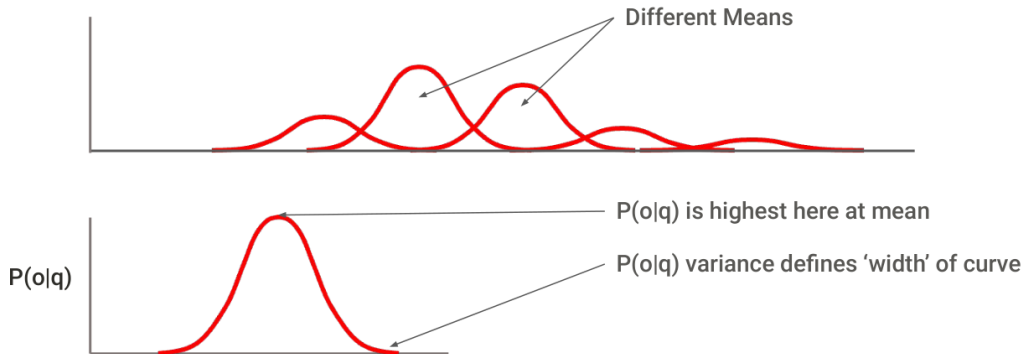


- **Mel Filter bank**

- Roughly uniformly spaced before 1 kHz
- logarithmic scale after 1 kHz

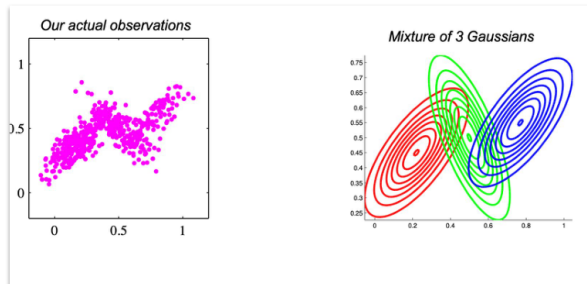


- $P(o|q)$: A single Gaussian parameterized by mean and variance. Mixture is a weighted sum.



Summary: each state has a likelihood function parameterized by:

- M Mixture weights
- M Mean Vectors of dimensionality D
- either:
 - M Covariance Matrices of $D \times D$
- or more likely:
 - M Diagonal Covariance Matrices of $D \times D$ equivalent to:
 M Variance Vectors of dimensionality D



- ▶ **Given our lexicon + HMM structure, and some acoustic model, we can:**
 - Generate the best alignment of HMM states to acoustic observations
- ▶ **With an alignment of HMM states to observations:**
 - Build a new acoustic model. Treat current state/obs mapping as training data+labels
 - This acoustic model is hopefully better than previous one
- ▶ **Repeat the align → rebuild acoustic model process until convergence**
 - Add parameters / complexity to acoustic model each iteration

Hybrid HMM-DNN ASR

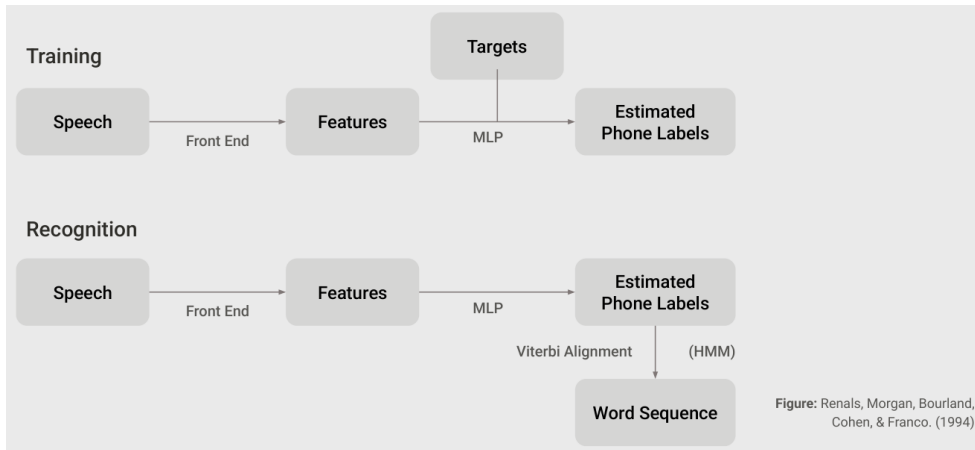
Transition to deep learning in acoustic modeling:
Hybrid HMM-DNN systems

- ▶ **Supervised learning**, minimize our classification errors
- ▶ **Standard choice: Cross entropy loss function**
 - Straightforward extension of logistic loss for binary

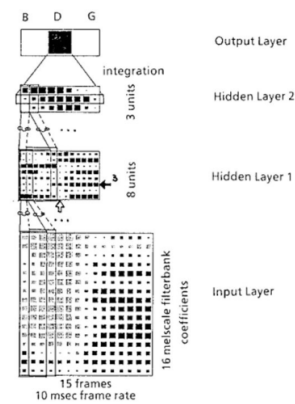
$$\text{Loss}(x, y; W, b) = - \sum_{k=1}^K \mathbb{I}(y = k) \log f(x)_k$$

- ▶ This is a **frame-wise** loss. We use a label for each frame from a forced alignment
- ▶ Other loss functions possible. Can get deeper integration with the HMM or word error rate

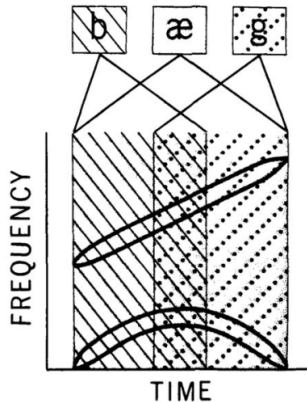
Neural nets for acoustic modeling goes back to early speech



Speech tasks motivated some of the original backpropagation work



Waibel, Hanazawa, Hinton, Shikano, & Lang. 1988



McClelland, & Elman. 1985

Hybrid systems took over ASR around 2012-2015

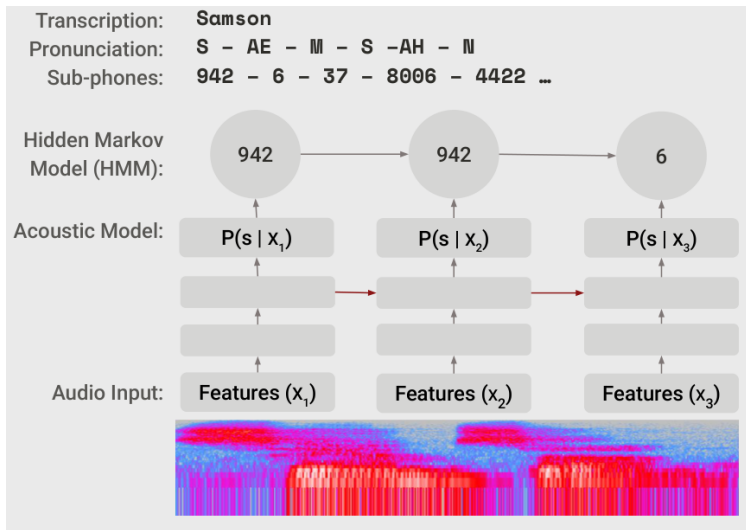
- ▶ DNN-HMM hybrids (Hinton et al., 2012) rapidly replaced GMM-HMM on large-vocabulary tasks.
- ▶ Sets the stage for deeper encoders, sequence training, and later end-to-end models.

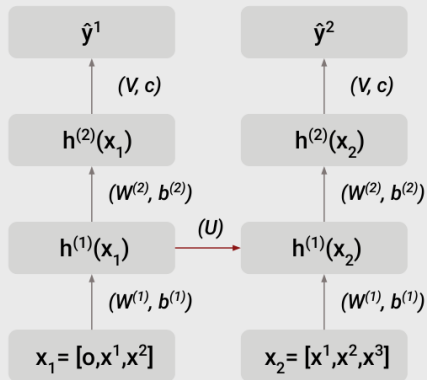
[TABLE 3] A COMPARISON OF THE PERCENTAGE WERs USING DNN-HMMs AND GMM-HMMs ON FIVE DIFFERENT LARGE VOCABULARY TASKS.

TASK	HOURS OF TRAINING DATA	DNN-HMM	GMM-HMM WITH SAME DATA	GMM-HMM WITH MORE DATA
SWITCHBOARD (TEST SET 1)	309	18.5	27.4	18.6 (2,000 H)
SWITCHBOARD (TEST SET 2)	309	16.1	23.6	17.1 (2,000 H)
ENGLISH BROADCAST NEWS	50	17.5	18.8	
BING VOICE SEARCH (SENTENCE ERROR RATES)	24	30.4	36.2	
GOOGLE VOICE INPUT	5,870	12.3		16.0 (>> 5,870 H)
YOUTUBE	1,400	47.6	52.3	

- ▶ Context-dependent HMM states
- ▶ Deeper nets improve on single hidden layer nets
- ▶ Hidden unit nonlinearity
- ▶ Many more model parameters (scaling up) & large datasets
- ▶ Specific depth (e.g. 3 vs 7 hidden layers)
- ▶ Fast computers = run many experiments
- ▶ Architecture choices (easiest is replacing sigmoid)
- ▶ Pre-training **does not matter***
 - Initially we thought this was the new trick that made things work

Recurrent DNN hybrid acoustic models





Output layer: $\hat{y}^2 = Vh^{(1)}(x_2) + c$

Hidden layer: $h^{(2)}(x_2) = \sigma(W^{(2)}h^{(1)}(x_2) + b^{(3)})$

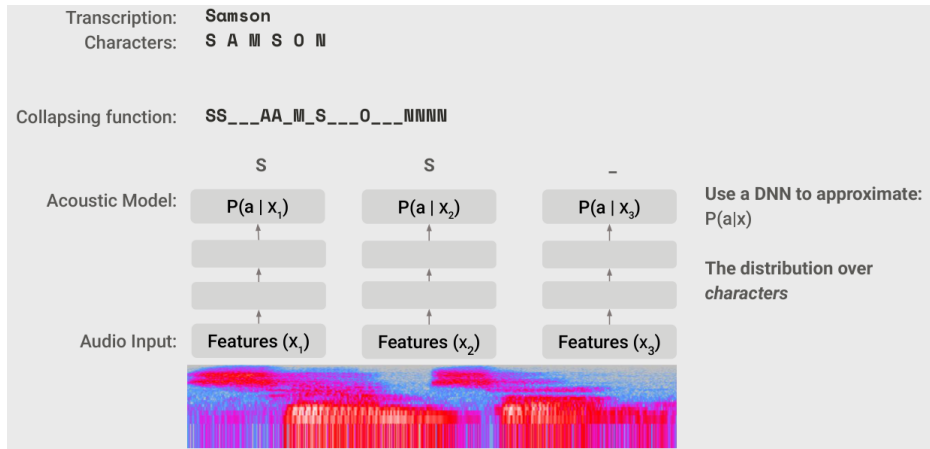
Hidden layer: $h^{(1)}(x_2) = \sigma(W^{(1)}x_2 + b^{(1)} + Uh^{(1)}(x_1))$

Input: x

- ▶ Lexicon introduces too many pronunciation assumptions. Requires hand engineering.
- ▶ Iteratively building HMM systems requires complex “recipes” to progressively improve alignments + acoustic models.
- ▶ HMM-DNN systems perform better, but still require the above
 - Can we use deep learning approaches to replace the HMM-based approaches so far?

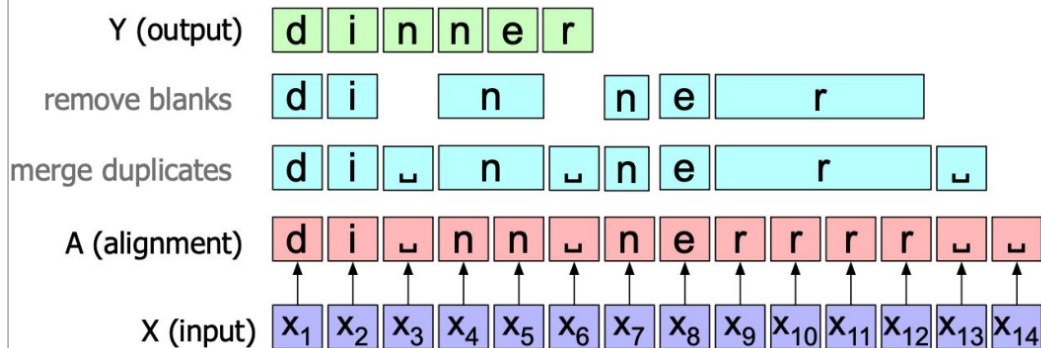
End-to-end alternative to hybrid HMM–DNN: map audio directly to transcript tokens (characters, phonemes, BPE).

- DNN approximates $P(a | x)$ over output tokens; no separate HMM state topology for alignment.



CTC output assumptions

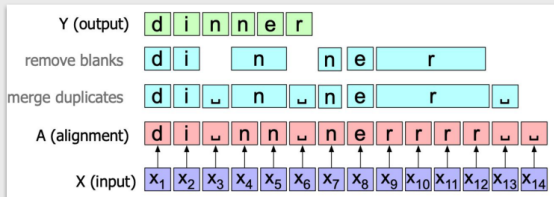
- ▶ Each timestep t the model outputs $P(a \mid X_{1:T})$, which can either be an output token (character, phoneme) or “blank”.
- ▶ *Assume:* Input sequence (length T) $\geq$ output sequence length (safe assumption in ASR).
- ▶ **Collapsing rule:** Ignore all blank tokens, and ignore repeated outputs of the same token.



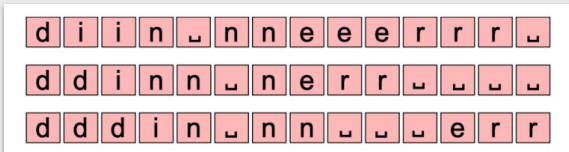
CTC collapsing. Sum over all possible alignments

Many alignments produce the same transcript output.

$P(\text{transcript}) = \text{sum over } P(\text{alignment}) \text{ for all alignments consistent with transcript}$



Additional alignments for 'dinner'



Real example. Full utterance with blank token

Per-frame argmax:

```
yy_ee      tt      a  
rr_e      hh      b ii      ll i      tt      aa      tt      iio_n  
cc      rrr_u      ii ss  
thh_e      o      nn      hhh_a      nnddd      i_n  
l o o g g ii nng      bb_uuii      llldd      ii      nng  
b_rr_ii      ck_s      p_ll_a      sstt      eerr  
a_nnd      b_ill_uu      ee_pp_r_i      nnss      f_oo_rrr_tt_y  
t_www_oo      nn_ew      b_e      t      i_n  
e      pp      aa_rr_tt      mm_ee      nntss
```

After collapsing:

YET A REHABILITATION CRU IS ONHAND IN THE BUILDING LOOGGING BRICKS PLASTER AND BLUEPRINS FOUR FORTY TWO
NEW BETIN EPARTMENTS

Reference

YET A REHABILITATION CREW IS ON HAND IN THE BUILDING LUGGING BRICKS PLASTER AND BLUEPRINTS FOR FORTY TWO
NEW BEDROOM APARTMENTS

Example CTC results. No word lexicon or LM (WSJ corpus)

YET A REHABILITATION CRU IS ONHAND IN THE BUILDING LOOGGING BRICKS PLASTER AND BLUEPRINS FOUR FORTY TWO
NEW BETIN EPARTMENTS

YET A REHABILITATION CREW IS ON HAND IN THE BUILDING LUGGING BRICKS PLASTER AND BLUEPRINTS FOR FORTY TWO
NEW BEDROOM APARTMENTS

THIS PARCLE GUNA COME BACK ON THIS ILAND SOM DAY SOO

THE SPARKLE GONNA COME BACK ON THIS ISLAND SOMEDAY SOON

TRADE REPRESENTIGD JUIDER WARANTS THAT THE U S WONT BACKCOFF ITS PUSH FOR TRADE BARIOR REDUCTIONS

TRADE REPRESENTATIVE YEUTTER WARNS THAT THE U S WONT BACK OFF ITS PUSH FOR TRADE BARRIER REDUCTIONS

TREASURY SECRETARY BAGER AT ROHIE WOS IN AUGGRAL PRESSED FOUR ARISE IN THE VALUE OF KOREAS CURRENCY

TREASURY SECRETARY BAKER AT ROH TAE WOO'S INAUGURAL PRESSED FOR A RISE IN THE VALUE OF KOREAS CURRENCY

- ▶ Labels at each time index are **conditionally independent** given the audio input (like HMMs):

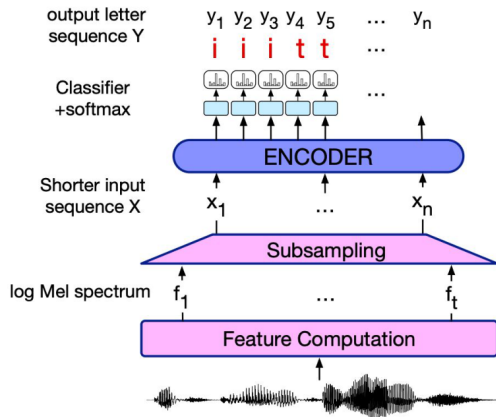
$$\Pr(a \mid x) = \prod_{t=1}^T \Pr(a_t \mid x)$$

- ▶ **CTC loss:** Negative log probability of the target transcription y^* , summed over all possible time-level labelings a that collapse to y^* :

$$\text{CTC}(x) = -\log \Pr(y^* \mid x), \quad \Pr(y \mid x) = \sum_{a \in B^{-1}(y)} \Pr(a \mid x)$$

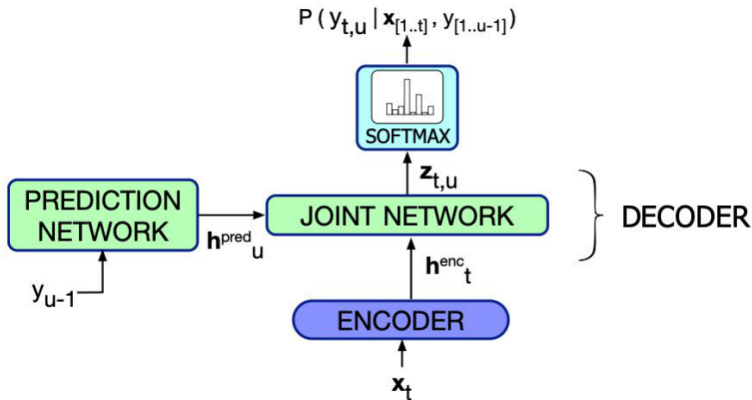
- ▶ For sequence length $T = 3$, transcript *HI*:
 - All possible pre-collapsed/time-level labelings: HHI, HII, H_I, _HI, HI_
- ▶ The final loss maximizes the probability of the transcript y^* .
- ▶ **Insights:** Conditional independence allows us to compute outputs for each t in parallel.
 - Blank + collapsing enables efficient, exact summation over all alignments.

- ▶ **Output model assumes conditional independence.**
- ▶ Each t most likely output depends only on inputs (not auto-regressive).
- ▶ Most likely pre-collapsed output is simply the sequence of argmax outputs for each t .
- ▶ Run CTC collapsing to obtain transcript.



Improving on CTC: RNN-Transducer (RNN-T) output model

- ▶ Add network that reasons over sequence of non-blank characters/tokens and input sequence
- ▶ Decoding outputs each transcript token only once. Optional blanks between. Variation on CTC
- ▶ Prediction network often pre-trained with CTC. Encoder provides transformed features for each t



Self-Supervised Learning: Wav2Vec 2.0

wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations

Alexei Baevski

Henry Zhou

Abdelrahman Mohamed

Michael Auli

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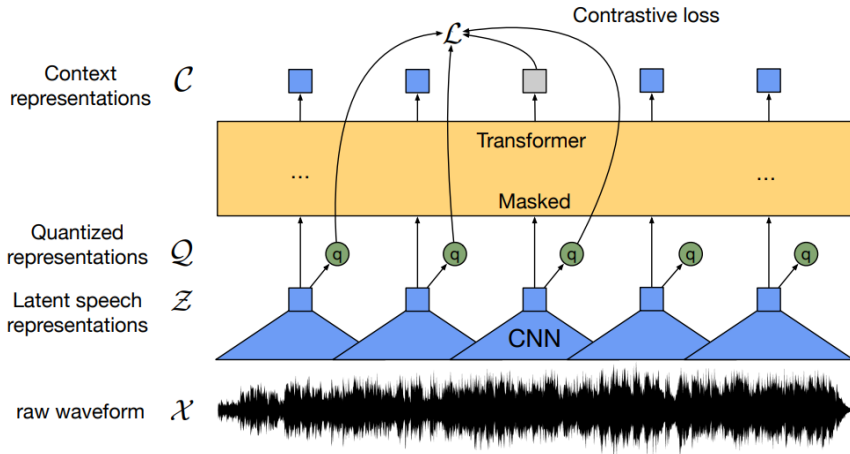
Facebook AI

Abstract

We show for the first time that learning powerful representations from speech audio alone followed by fine-tuning on transcribed speech can outperform the best semi-supervised methods while being conceptually simpler. wav2vec 2.0 masks the speech input in the latent space and solves a contrastive task defined over a quantization of the latent representations which are jointly learned. Experiments using all labeled data of Librispeech achieve 1.8/3.3 WER on the clean/other test sets. When lowering the amount of labeled data to one hour, wav2vec 2.0 outperforms the previous state of the art on the 100 hour subset while using 100 times less labeled data. Using just ten minutes of labeled data and pre-training on 53k hours of unlabeled data still achieves 4.8/8.2 WER. This demonstrates the feasibility of speech recognition with limited amounts of labeled data.¹

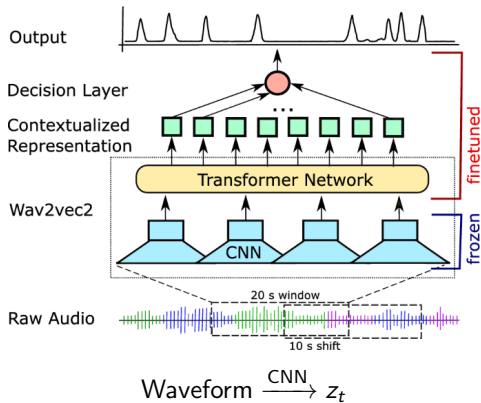
- ▶ **Proposed by Baevski et al. (2020)**
- ▶ **Key Idea:** Self-supervised learning for speech representation.
- ▶ **Architecture:**
 - CNN encoder for raw waveform $\rightarrow$ latent representations z_t
 - Transformer for context representations c_t
 - Quantization module for discrete representations q_t

Wav2Vec 2.0: Overview (cont.)



Wav2Vec 2.0 architecture overview.

- **Step 1:** Raw audio waveform is input to a multi-layer CNN encoder.
- **Output:** Latent speech representations z_t .



- ▶ **Step 2:** Random spans of z_t are masked.
- ▶ **Step 3:** Transformer network produces contextualized representations c_t .

$$z_t \xrightarrow{\text{mask spans}} \xrightarrow{\text{Transformer}} c_t$$

- ▶ **Step 4:** Quantize z_t to discrete representations q_t .
- ▶ **Quantization:** Product of G codebooks, each of size V .

$$z_t \xrightarrow{\text{Quantization}} q_t$$

- **Contrastive Loss:** Distinguish true quantized vector q_t^+ from negatives q^- .

$$L_t = -\log \frac{\exp(\text{sim}(c_t, q_t^+))}{\sum_{q^-} \exp(\text{sim}(c_t, q^-))}$$

- **Diversity Loss:** Encourages usage of all codebook entries.

► State-of-the-art results:

- $\sim 1.8\%$ WER with 960h labeled data
- $\sim 4.8\%$ WER with only 10 minutes labeled data

► **Impact:** Enables efficient use of unlabeled speech data.

TIMIT phoneme recognition accuracy in terms of phoneme error rate (PER).

	dev PER	test PER
CNN + TD-filterbanks [59]	15.6	18.0
PASE+ [47]	-	17.2
Li-GRU + fMLLR [46]	-	14.9
wav2vec [49]	12.9	14.7
vq-wav2vec [5]	9.6	11.6
This work (no LM)		
LARGE (LS-960)	7.4	8.3

- ▶ After CTC/RNN-T hybrids, **Conformer** encoders and **Whisper**-scale weak supervision define current practice.

Conformer: Convolution-augmented Transformer for Speech Recognition

- ▶ Sequence-to-sequence transformer with multi-headed self attention.
- ▶ Combines attention (global context) with convolution (local invariance).
- ▶ Uses RNN-T loss architecture.

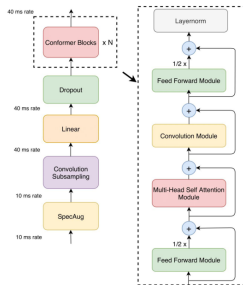
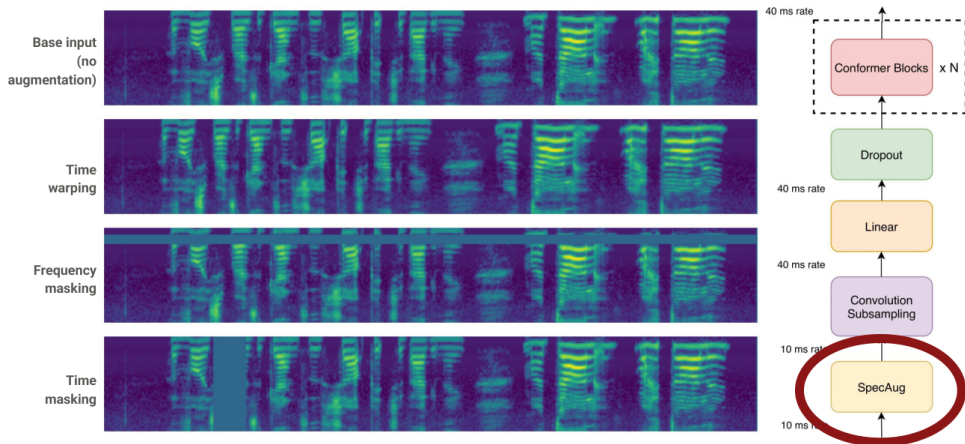


Figure: Conformer encoder model architecture. Comprises two macaron-like feed-forward layers with half-step residual connections sandwiching the multi-headed self-attention and convolution modules. This is followed by a post layernorm. (Gulati *et al.*, 2020)

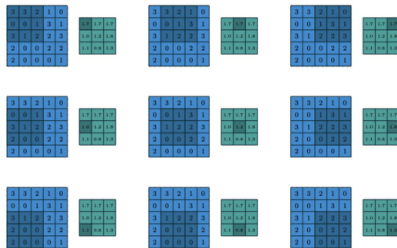
SpecAugment: Augmenting data to increase dataset size

- Mask time/frequency bands in log-mel during training for robustness.

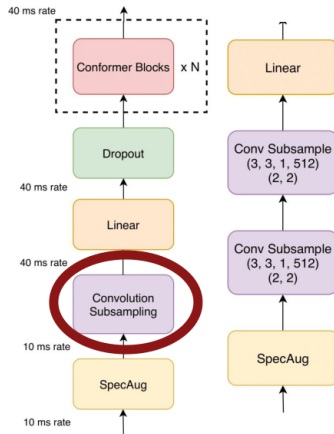
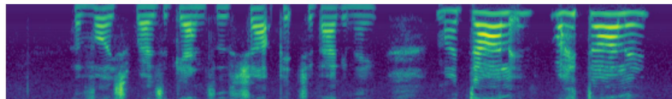


[Park et al., 2019](#)

Convolution subsampling: Log mel spectrogram is an image



[Conv image source](#)



The convolution block has the greatest effect on performance

- ▶ From ablation experiment of the paper, **removing unique features and moving towards a vanilla Transformer block** was tested.
- ▶ All ablation study results are evaluated without external LM.

The convolution block has the greatest effect on performance.

Model Architecture	dev clean	dev other	test clean	test other
Conformer Model	1.9	4.4	2.1	4.3
– SWISH + ReLU	1.9	4.4	2.0	4.5
– Convolution Block	2.1	4.8	2.1	4.9
– Macaron FFN	2.1	5.1	2.1	5.0
– Relative Pos. Emb.	2.3	5.8	2.4	5.6

Ablation results: Largest WER drop from convolutions.

Multitask training data (680k hours)

English transcription



"Ask not what your country can do for ..."



Ask not what your country can do for ...

Any-to-English speech translation



"El rápido zorro marrón salta sobre ..."



The quick brown fox jumps over ...

Non-English transcription



"언덕 위에 올라 내려다보면 너무나 넓고 넓은 ..."



언덕 위에 올라 내려다보면 너무나 넓고 넓은 ...

No speech



(background music playing)



∅

([Radford et al.](#), 2022)



Ground truth: Comme moi, vous avez peut-être déjà vécu cette scène. Qu'est-ce que tu fais cet été toi? Écoute, je ne sais pas encore trop.

Whisper (French): Comme moi, vous avez peut-être déjà vécu cette scène. Qu'est-ce que tu fais cet été toi? Écoute, je ne sais pas encore trop.

Whisper (Translation): Like me, you may have already experienced this scene. What are you doing this summer? I don't know yet

Whisper: How does it perform a variety of tasks accurately?

Multitask training data (680k hours)

English transcription

- 🗣️ "Ask not what your country can do for ..."
- 📄 Ask not what your country can do for ...

Any-to-English speech translation

- 🗣️ "El rápido zorro marrón salta sobre ..."
- 📄 The quick brown fox jumps over ...

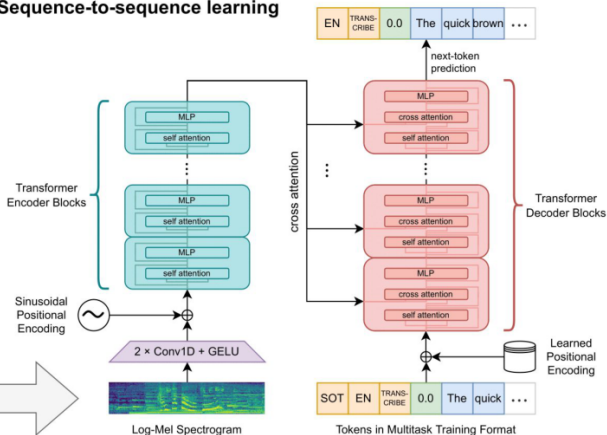
Non-English transcription

- 🗣️ "언덕 위에 올라 내려다보면 너무나 넓고 넓은 ..."
- 📄 언덕 위에 올라 내려다보면 너무나 넓고 넓은 ...

No speech

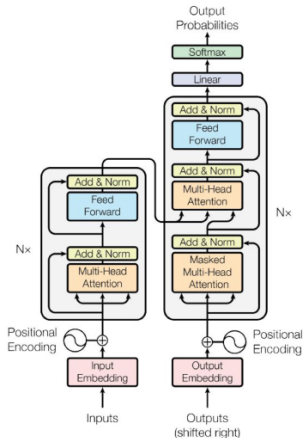
- 🔊 (background music playing)
- 📄 ∅

Sequence-to-sequence learning



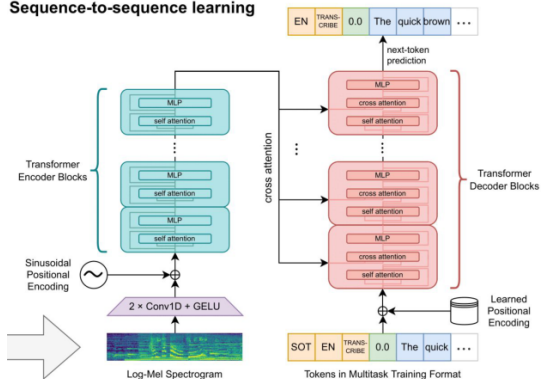
(Radford et al. 2022)

Text encoder-decoder transformer



Whisper encoder-decoder transformer

Sequence-to-sequence learning



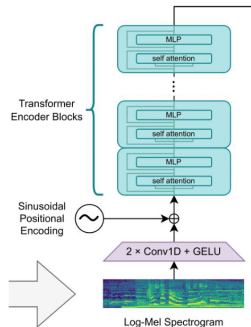
Whisper encoder:

transformer encoder

- ▶ Encoder input: log-mel spectrogram and two convolutional layers
- ▶ Positional embeddings
- ▶ Standard transformer blocks depending on model size

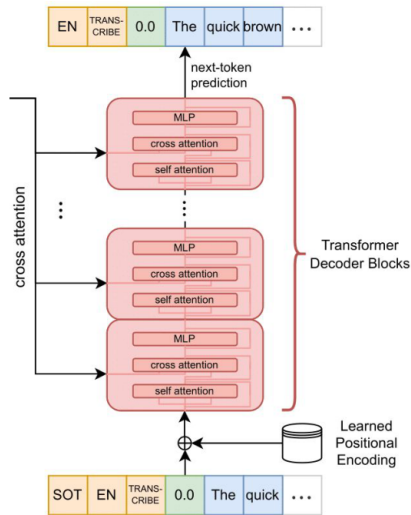
CS 224S / LINGUIST 285

Lecture 11: State-of-the-art deep learning approaches for speech recognition.

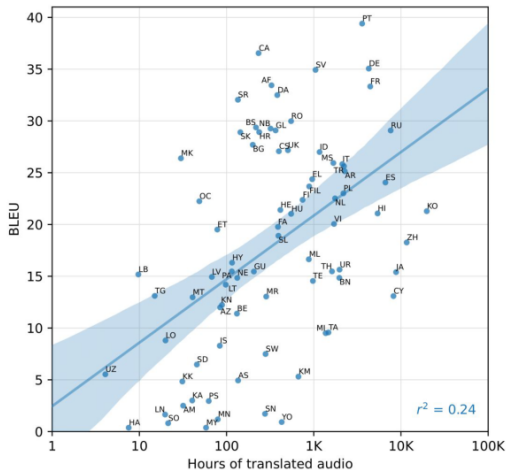


Whisper decoder: autoregressive transformer decoder

- ▶ **GPT-2 style** model: no more CTC loss.
- ▶ Decodes using **auto-regressive prediction**—each token depends on previous (text prediction).
- ▶ **Cross-attention:** To encoded audio representation from encoder.
- ▶ **Custom vocabulary and tags:**
 - Language tags: <en>, <fr>, etc.
 - Task tags: <transcribe>, <translate>
 - **Timestamps**
 - Speech tags: <nospeech>



[illegible]



Speech translation performance on FLEURS dataset (higher is better)

- ▶ Translation performance is better on average after the 100 hours of training data mark.
- ▶ Amount of training data isn't as correlated with translation performance as it is with transcription performance.
- ▶ Although possible, speech translation is likely to only be reliable for high resource languages.

Streaming ASR

Transcribing while the user is still speaking

- ▶ **Offline (full-context):** the whole utterance is available before decoding starts. Best accuracy; fine for transcribing recordings.
- ▶ **Streaming:** tokens must appear while audio is still arriving. Required for voice assistants, live captions, and dictation.
- ▶ The core constraint: a streaming model may only use the **past** (and a small fixed lookahead), never the whole future of the utterance.
- ▶ User-perceived latency comes from three places: acoustic lookahead, model computation, and the decision that the speaker has finished.

- ▶ Full self-attention encoders (Conformer, Whisper) attend over the entire utterance, so they cannot stream as-is.
- ▶ **Causal encoders:** restrict attention and convolutions to the past only. Cheapest latency, largest accuracy loss.
- ▶ **Chunked attention:** process audio in fixed chunks with a small right-context window; each chunk sees a little of the future.
- ▶ Chunk size is the latency-accuracy knob: bigger chunks recover offline accuracy, smaller chunks respond faster.

- ▶ The RNN-Transducer is **frame-synchronous**: at each audio frame it either emits a token or a blank and moves on, so partial transcripts appear as audio arrives.
- ▶ No attention over the full input and no need to wait for the end of the utterance, unlike encoder-decoder models with cross-attention.
- ▶ The prediction network conditions on previously emitted tokens, giving language-model-like context without breaking streamability.
- ▶ Production systems (mobile dictation, on-device assistants) are overwhelmingly RNN-T variants with causal or chunked encoders.

- ▶ **Endpointing** is deciding that the speaker has finished, so the final result can be committed and the microphone closed.
- ▶ Too eager: the system cuts users off mid-sentence. Too patient: every interaction feels slow.
- ▶ Signals used: silence duration, the decoder's end-of-utterance probability, and prosodic cues.
- ▶ Endpointing errors often dominate perceived quality even when word accuracy is high.

► How many hours of data do you have?

- If you have hundreds of hours of data and need speech recognition for a specific domain, you could get away with training your own model or fine tuning an existing model.
- If you have tens of hours of data, you can finetune an existing model.

► Which language are you looking to transcribe or translate?

- If the model is well represented in Whisper, it is a strong baseline, especially for translation into English.
- If you'd like to translate into another language, you will need your own model.

► What is your compute budget?

- If you don't have any compute budget, using the best model you can find may be the best way forward.
- If you have a small compute budget, finetuning a small model variant could outperform a larger model out-of-the-box.
- If you have a large compute budget (and large enough dataset) you can consider finetuning a larger model variant.

- ▶ Noise, codecs, mic quality, overlap, and domain mismatch dominate field WER.
- ▶ Demographic bias in accents and dialects—audit before justice, hiring, or grading use cases.
- ▶ Voice is biometric: consent, retention, on-device options, and transparent API error rates.

- ▶ **Datasets:** LibriSpeech, GigaSpeech, Common Voice, AMI, CHiME, Fleurs—match to deployment domain.
- ▶ **Toolkits:** ESPnet, NeMo/Riva, Fairseq, SpeechBrain, Hugging Face transformers.



1. Stanford CS224S (2025): [lec9](#) (classical ASR), [lec10](#) (neural acoustic modeling), [lec11](#) (Conformer, SpecAugment, Whisper).
2. Edinburgh ASR course (2025): [lectures-2025](#).
3. CMU 11-751/11-752 speech recognition: [WAVLab](#).
4. JHU CLSP intro HLT slides; MIT 6.345/6.541J lecture notes (signal production, perception).
5. Graves et al., CTC, ICML 2006: [icml_2006.pdf](#).
6. Gulati et al., Conformer, Interspeech 2020: [arXiv:2005.08100](#).
7. Radford et al., Whisper, 2022: [arXiv:2212.04356](#).
8. Baevski et al., wav2vec 2.0, 2020: [arXiv:2006.11477](#).
9. Chan et al., Listen Attend and Spell, 2015: [arXiv:1508.01211](#).
10. Graves, Sequence Transduction (RNN-T), 2012: [arXiv:1211.3711](#).

11. Hinton et al., DNN acoustic modeling, 2012: [DNN-2012-proof.pdf](#).
12. Mohri et al., WFSTs in speech recognition; Distill CTC; Hugging Face ASR evaluation docs [47].